

# Introduction to Data Processing

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# Who am I?

- Full Professor, Université Côte d'Azur, Polytech Nice Sophia
- Member of I3S (computer science labs), team “Signals, Images and Systems”
- Research interests: data science, machine learning, deep learning, statistics, signal and image processing, biological data processing.
- Head of a Data Science engineer's degree at Engineering School “Polytech Nice Sophia”



# Staff

- Lectures
  - Lionel Fillatre
- Labs
  - Vanessa Lopez-Norena
  - Florin Baumann
  - Lionel Fillatre

# Class website

- Class website

<https://lms.univ-cotedazur.fr/2025/course/view.php?id=4156>

# Lecture 1

Introduction

# Outlines

1. Examination
2. Kaggle
3. Where to find data?
4. Data Science
5. Conclusion

# Examination

# Timeline of this course

1. Business problem and data *(01/06/2026 morning)*
2. Handling data *(01/06/2026 afternoon)*
3. Data visualization and data storytelling *(02/06/2026 morning)*
4. Data pipeline for feature engineering and linear/logistic/softmax regression *(02/06/2026 afternoon)*
5. Feature engineering and visualization *(03/06/2026 morning)*
6. Linear regression and storytelling *(03/06/2026 afternoon)*
7. Final data pipeline and oral defence preparation *(04/06/2026 morning)*
8. **Oral defence** and **report writing** *(05/06/2026 morning)*

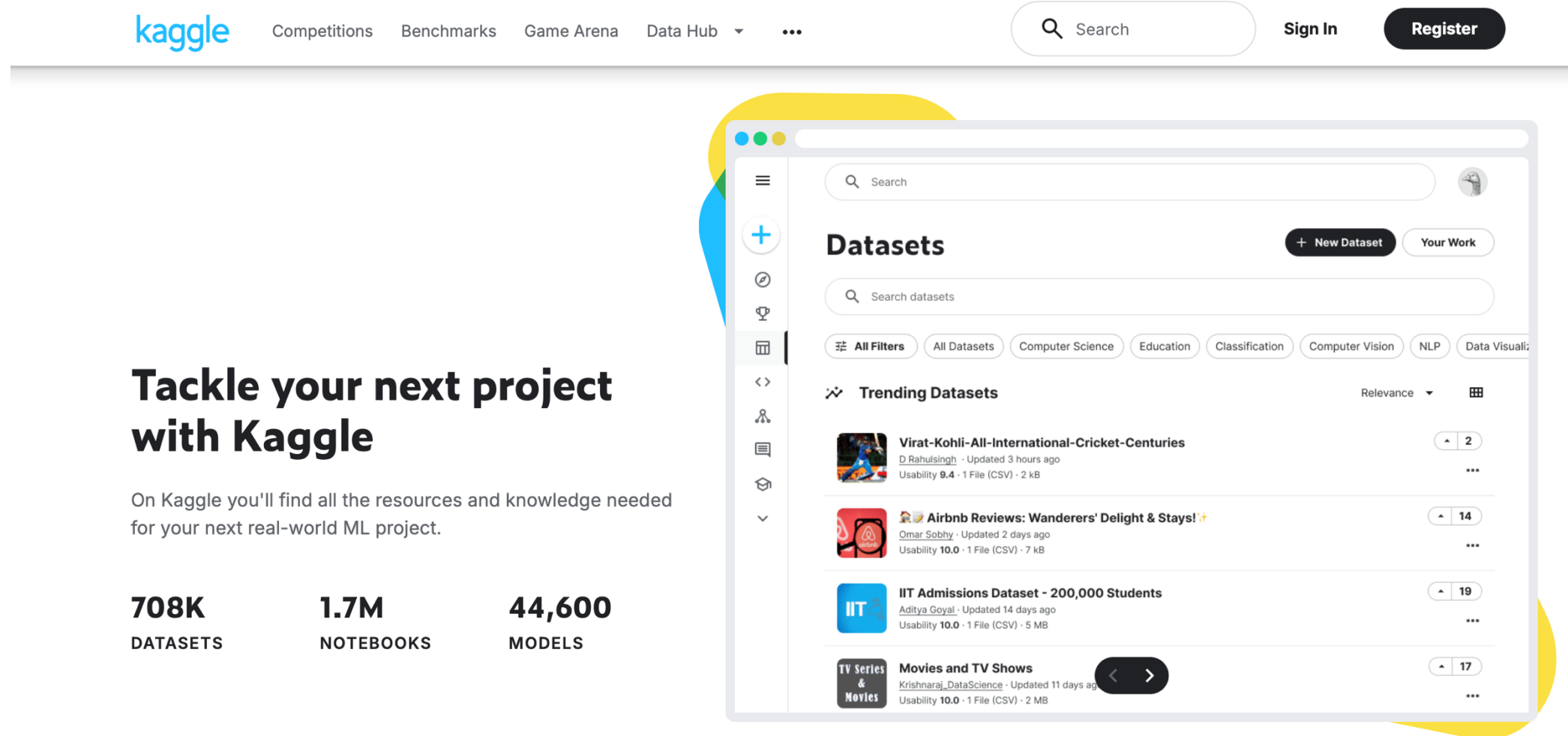


# Kaggle

# Kaggle

- Kaggle is a platform for predictive modelling and analytics competitions
- Companies and researchers post their data
- Statisticians, data miners, data scientists (and others) from all over the world compete to produce the best models.
- Website: <https://www.kaggle.com/>

# Kaggle website



# Kaggle website

- **Competitions**
  - The competition host prepares the data and a description of the problem
- **Datasets**
  - With or without competition
- **Codes**
  - Contain both the code needed for an analysis, and the analysis itself. It's the core of a work, what it needs to make it reproducible, to make it grow, and to invite collaboration.
- **Discussion:** forum of discussions
- **Jobs:** Hiring? Seeking?
- **Learn:** learn the basics to confidently start a new career or upgrade your skills.
- **Blog:** official blog of Kaggle.com
- **User rankings:** ranking of Kaggle users
- **Tags:** to find pages associated to a specific tags
- **Host a competition:** Kaggle can help you solve difficult problems, recruit strong teams, and amplify the power of the data science talent.

# Kaggle Challenge

- Work in group of 3 students
- Teams are chosen randomly
- You have to choose one dataset (**each team must have its own dataset**; no duplication; Titanic dataset is not allowed)
- Choose the dataset according to your tastes

# Kaggle Challenge

- You have to define a business question you aim to answer.
- You do not have to answer it perfectly! The data processing should shed light on this business question.
- You have to provide a data analysis to prepare the answer!
- I encourage you to reuse existing Kaggle notebooks
- You can write, run, and view best practice code and visualizations of this dataset.

# Git

- You are required to use a Git repository.
- Create and use the Git repository starting now.
- Provide the Git repository URL in your report.



# Deliveries of the challenge

- Oral defence (**5 minutes for the team + 5 minutes for questions and answers**):
  - You must promote your data challenge
  - The notebook is not required
  - The slides should be uploaded on Moodle on Thursday evening
  - **Required content (5 slides , one slide for each topic):**
    1. **Business goal:** what is your dataset? why is this dataset useful?
    2. **Data description:** how many features? Types? Etc.
    3. **Handcrafted features** describe and justify your handcrafted features.
    4. **Regression:** Use a linear/logistic/softmax regression to solve your problem.
    5. **Conclusion:** what are your benefits of this data analysis? Is it possible to improve your results? If so, how? If not, why?
- **A “data storytelling” format is required:** adopt a “data storytelling” style by turning your challenge and data into a clear and engaging narrative.



# Written report (10 pages maximum)

- **Required content:**

1. **Business goal (1 page max.):** what is your dataset? why is this dataset useful?
2. **Team management (1 page max.):** how do you work? What is your schedule?
3. **Data visualisation (2 pages max.):** propose some very good figures and describe them carefully.
4. **Handcrafted features (2 pages max.):** what are your handcrafted features? How do you derive them? How do you check out their relevance?
5. **Linear/logistic/softmax regression (2 pages max.):** Which kind of regression did you choose? Why? Are the results reliable?
6. **Conclusion (1 page max.):** what are your benefits of this data analysis? is it possible to improve your results? If so, how? If not, why?
7. **List of references (1 page max.):** what external sources of information did you use? Be sure to reference these sources throughout your report.

# Where to find data?

# Where to find data?

- Open Data
  - Open data is freely available on the internet. Any user is permitted to download, copy, analyse, re-process, and re-use it for any other purpose with minimal financial, legal, and technical barriers.
- The benefits of using an open dataset include (but are not limited to):
  - Access to the data is **free** and usually **fast** (without requiring a registration/approval process).
  - Other researchers may have used the same dataset and published guidelines, code and other details about the dataset that are also in the **public domain**. So we can benefit and draw inspiration from their **prior work**.
  - **Transparency & reproducibility**: If the dataset we're using is open we can make all our research (data, code & publications) open as well, so others can easily reproduce and contribute to our work.

# Sources of Open Data

- Tailored for Data Science
  - Some datasets have been heavily used for data science applications, and are available in easy-to-use formats that may already be cleaned/pre-processed for you.
  - Many popular machine learning libraries like PyTorch have datasets built-in, for example. These are excellent sources for finding datasets to learn, test and benchmark algorithms or other analysis techniques, but are less likely to be rich sources for novel research projects.
  - Main sources: Kaggle, UCI Machine Learning Repository, Google research, scikit-learn, Tensorflow, PyTorch
- Countries/Governments
  - Many governments commit to publishing data in the open for public interest and transparency. The datasets might be less “data science ready”, but they cover a broad range of topics.
  - UK: data.gov.uk, UK Data Service, Office for National Statistics (ONS)
  - Other countries/regions: data.gov (USA), data.europa.eu (EU), africaopendata.org (Africa), dataportal.asia (Asia)
- Organisations
  - Large humanitarian organisations often make data available, such as World Bank, World Health Organisation, International Monetary Fund

# Legality and Ethics

- Commercial agreements
  - Research projects are often collaborative efforts that may span multiple organisations.
  - Commonly, these collaborations take place under some defined commercial agreement (a contract).
  - Such an agreement may impact on your use of data.
- Ethics
  - Data ethics encompasses the moral obligations of gathering, protecting, and using personally identifiable information and how it affects individuals.

# Legality and Ethics

- Data Protection & General Data Protection Regulation (GDPR).
  - Everyone responsible for using personal data has to follow strict rules called ‘data protection principles’.
  - They must make sure the information is:
    - used fairly, lawfully and transparently.
    - used for specified, explicit purposes.
    - used in a way that is adequate, relevant, and limited to only what is necessarily accurate and, where necessary, kept up to date.
    - kept for no longer than is necessary.
    - handled in a way that ensures appropriate security, including protection against unlawful or unauthorised processing, access, loss, destruction, or damage.

# Data Sources and Formats

- Tabular Data Files
  - Comma-separated values files (CSVs)
    - Tabular data files, and particularly comma-separated values (CSV) files, are likely to be the data format you encounter most often.

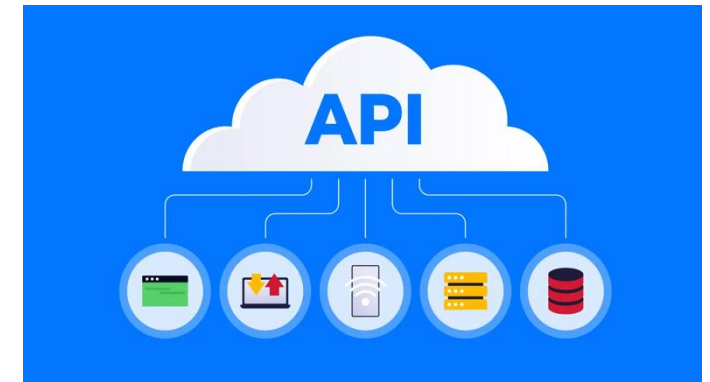
```
"Subject","Teacher","Day of Week","Time Start","Time End"  
"Maths","Mr F","Monday",1000,1200  
"English","Ms P","Tuesday",1100,1300  
"Physics","Mrs A","Thursday",1400,1600  
"French","Mr F","Friday",1000,1300
```

- Excel Spreadsheets
  - As CSV files are a plaintext (human readable) format and don't need proprietary software to use and create they are always preferable to Excel spreadsheets for raw data, if available. However, it is still common to find data in Excel .xls or .xlsx files.

# Data Sources and Formats



- Databases
  - We can think of databases as containers for data.
  - Typically, databases allow for easy storage, retrieval, update, and analysis of data.
  - In the case of tabular data, a database may have many tables.
  - A database can be flat, or it can be relational.
  - Two main types:
    - SQL Databases & RDBMS
    - NoSQL Databases
- Application Programming Interfaces (APIs)
  - Application programming interfaces (APIs) are services that allow you to programmatically request information, including datasets, from a remote server.





# Data Types



- Image Data
  - Images are commonly represented as a  $n$ -dimensional tensor of pixel values.
- Text Data
  - Text data is common in data science applications, anything from analysing Tweets and social media to documents and novels.

# Data Types

- Other types
  - **Audio:** commonly stored as .wav or .mp3 and displayed as a waveform - digitised audio can be used for things like training speech recognition models. Lots of crossover with signal processing! Librosa is a popular python package for working with audio.
  - **Video:** can be thought of as many images + audio!
  - **Geospatial:** data relating to any location on the Earth's surface. The geopandas library combines the capabilities of shapely and Pandas to make working with this datatype easier.
  - **Time Series:** Data that can be expressed as observations over time, e.g., stock price data. Honourable mention to sktime for analysis.
  - **XML (Extensible Markup Language):** using a library such as BeautifulSoup can help parse this heavily structured format.
  - **Web scraping:** extracting structured data from web sites. Scrapy is a popular library here.

# Pandas

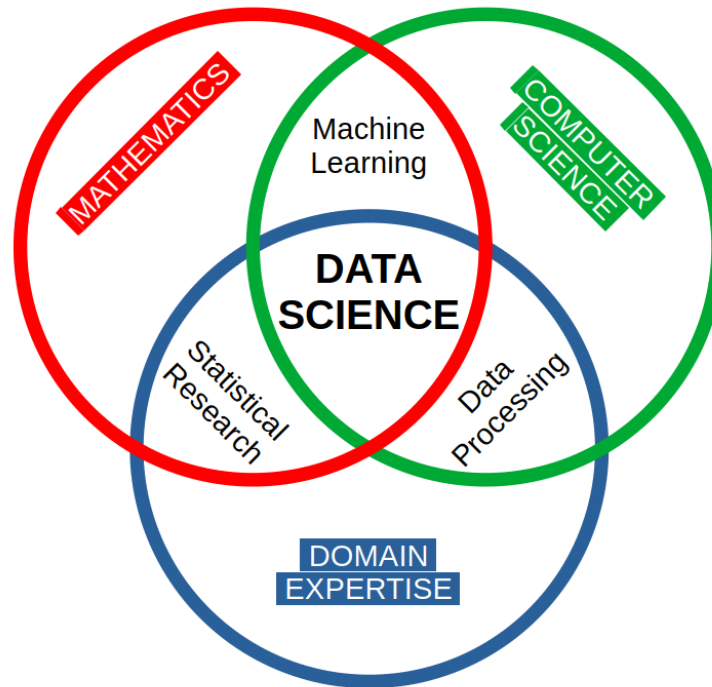


- The Pandas library is a core part of the Python data science ecosystem. It provides easy-to-use data structures and data analysis tools.
- Pandas has some great resources for getting started, including guides tailored to those familiar with other software for manipulating data:  
<https://pandas.pydata.org/pandas-docs/stable/index.html>

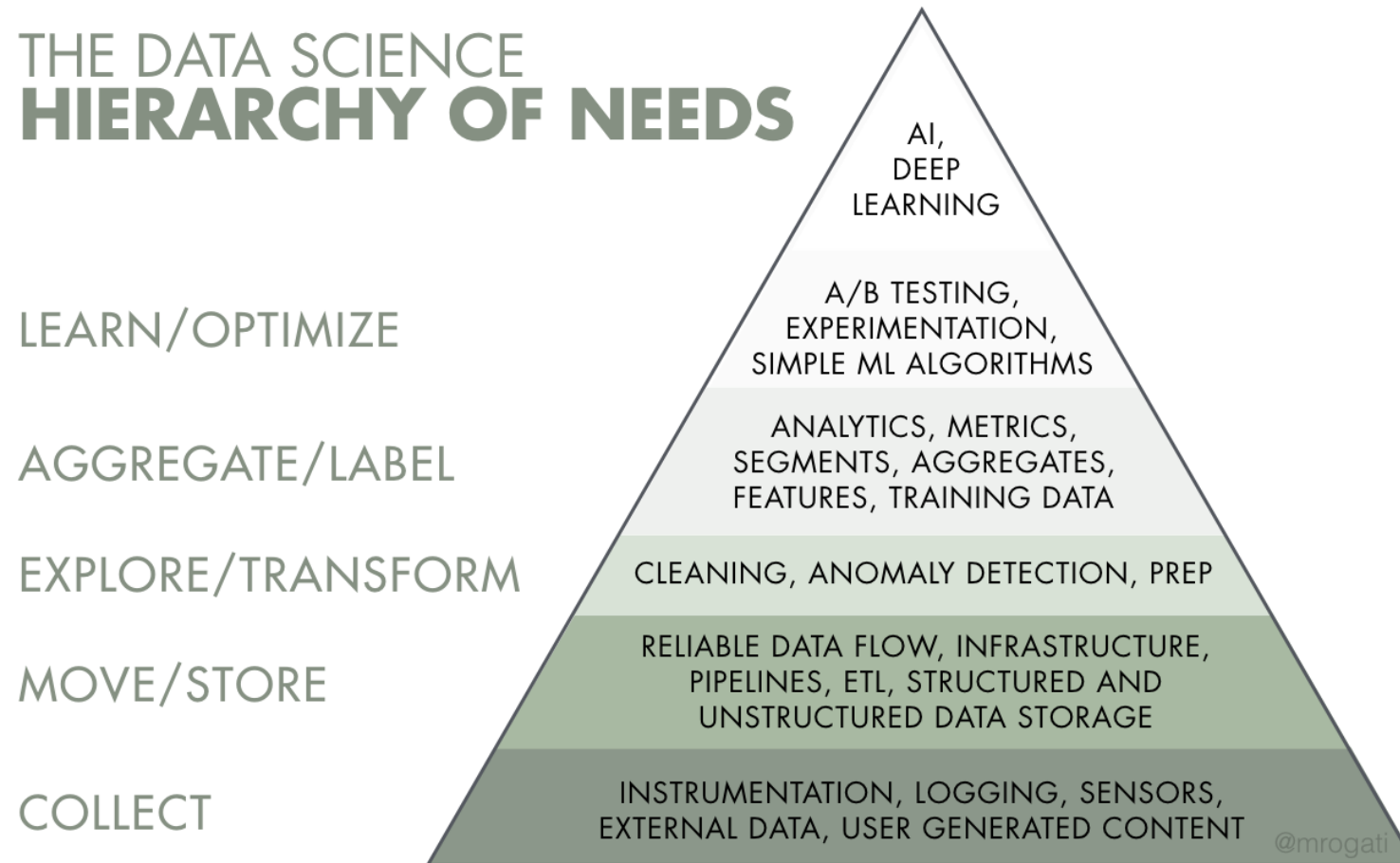
# Data Science

# Data Science

- The term **Data Science** refers to a varied ensemble of practices, methodologies, and tools that may be used to learn from or about data.



# Monica Rogati's representation of the "Data Science hierarchy of needs"



# Data

- The availability of large-scale datasets, for example for training face recognition algorithms or language models, is a fairly new phenomenon.
- Creating, refining and, making data consumable takes a substantial effort, and acquiring data remains complex (e.g., digitisation).
- These processes should not be taken for granted. Two important considerations to keep in mind when faced with a dataset are **how the data was created**, and **how representative the data is**.

# Data Creation (1/2)

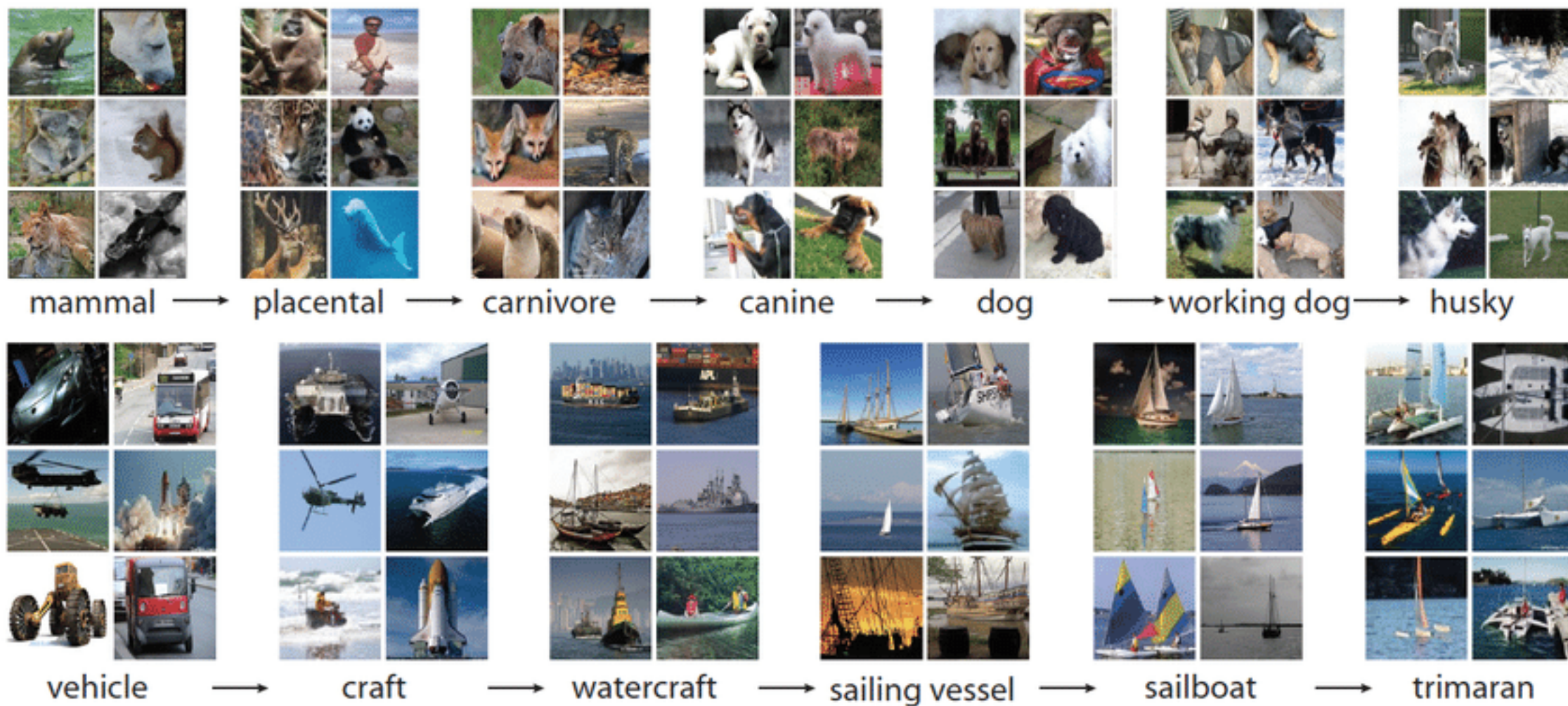
- Data is not a natural resource, it is always the product of human decisions and actions.
- Data is often the outcome of an enormous amount of labour, resources, infrastructure, and time.
- Example data creation tasks are:
  - collecting information or the tracking of historical information
  - organising information in specific categories
  - measuring and storing information as data on digital infrastructure
- Whenever a dataset is enriched with metadata information or specific labels, keep in mind someone has provided those.



# Data Creation (2/2)

- For example, the famous ImageNet dataset, central component for the development of many well known image recognition pipelines relies on two pillars:
  1. a taxonomy developed since 1985 as part of the lexical database WordNet, which provides a top-down hierarchical structure of concepts
    - Example: “chair” is under artifact->furnishing->furniture->seat->chair
  2. an enormous amount of cheap workforce provided by Amazon Mechanical Turk.
- Amazon Mechanical Turk
  - Employers, known as requesters, post jobs known as Human Intelligence Tasks (HITs), such as identifying specific content in an image or video, writing product descriptions, or answering survey questions.
  - Workers, colloquially known as Turkers or crowdworkers, browse among existing jobs and complete them in exchange for a fee set by the requester.

# Imagenet





# Imagenet

## Geological formation, formation

(geology) the geological features of the earth

1808  
pictures

86.24%  
Popularity  
Percentile



Numbers in brackets: (the number of synsets in the subtree).

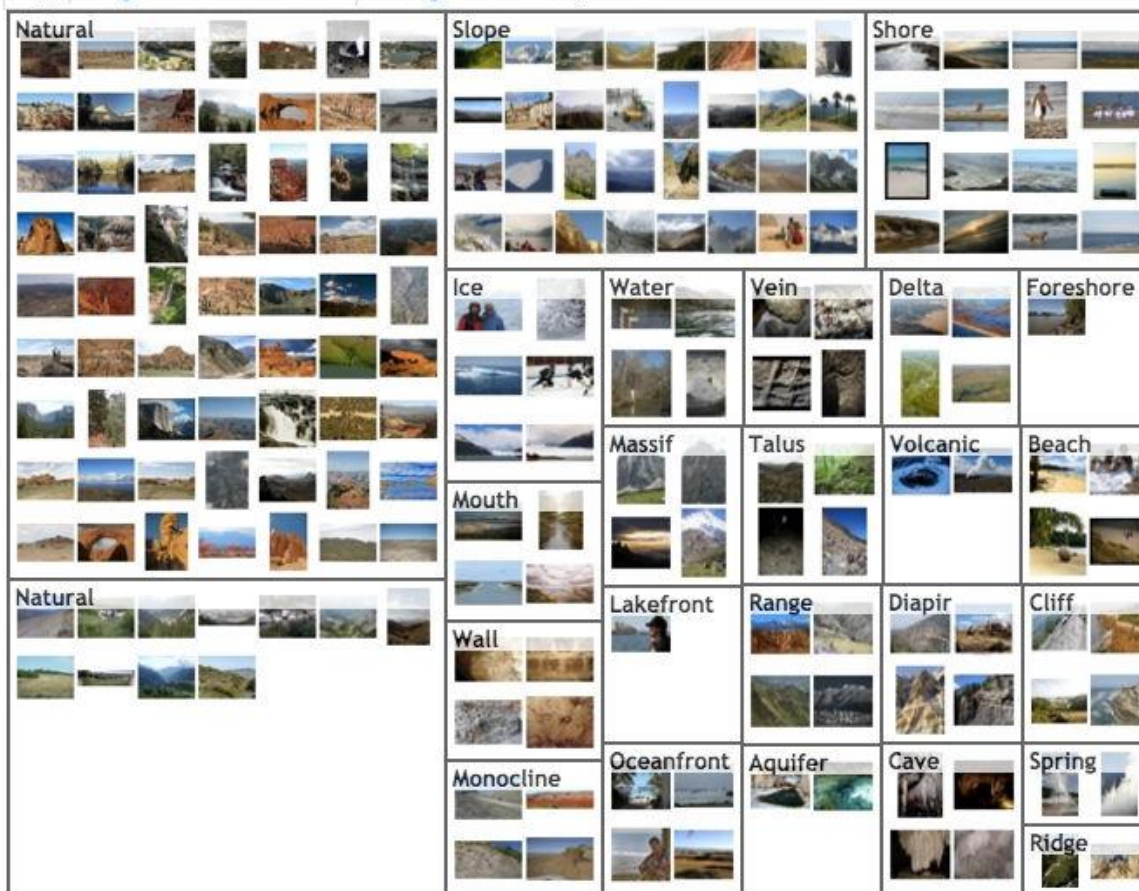
- ImageNet 2011 Fall Release (32326)
  - plant, flora, plant life (4486)
  - geological formation, formation (1808)
    - aquifer (0)
    - beach (1)
    - cave (3)
    - cliff, drop, drop-off (2)
    - delta (0)
    - diapir (0)
    - folium (0)
    - foreshore (0)
    - ice mass (10)
    - lakefront (0)
    - massif (0)
    - monocline (0)
    - mouth (0)
    - natural depression, depression (0)
    - natural elevation, elevation (41)
    - oceanfront (0)
    - range, mountain range, range of mountains (0)
    - relict (0)
    - ridge, ridgeline (2)
    - ridge (0)
    - shore (7)
    - slope, incline, side (17)
    - spring, fountain, outflow, outpouring (0)
    - talus, scree (0)
    - vein, mineral vein (1)
    - volcanic crater, crater (2)
    - wall (0)

### Treemap Visualization

### Images of the Synset

### Downloads

ImageNet 2011 Fall Release Geological formation, formation



# Data Representativeness

- The data we have available is just a sample and not the complete story.
- The question we should ask ourselves is:  
“What can these data points tell us about the wider phenomenon that we’re really interested in?”
- Anything less than the whole data universe is a sample.
- Given the existing social structures, no matter how big that sample is, it is not representative due to (amongst other things)
  - unequal access to technology,
  - unequal possibility to defend one’s privacy and copyright,
  - and limited access to the huge volumes of speech (for example) produced in the “walled garden” platforms like Facebook.

# Collection and Storage

- Part of the work of a data scientist is knowing the challenges and hurdles involved in data collection and storage.
- It is essential that we know who owns the data, what restrictions apply, how a resource should be stored for long-term preservation and made available to collaborators.
- In small team contexts' data scientists often take care directly of such responsibilities.

# Wrangling

- Commonly, we need to alter a freshly obtained dataset to a more analysis-ready state.
- We might need to remove inconsistencies and inaccuracies (a process called **data cleaning**), combine data sources, or otherwise change the data into our desired format.
- These tasks can be collectively called **data wrangling**.
- Although often this task is the largest part of a data scientist's work it is often undervalued.

# Conclusion

# Next

- Choose your Kaggle dataset
- Define your business problem
- Check that your dataset is representative for your problem